GENERATIVE ADVERSARIAL NETWORKS

Generative Artificial Intelligence
M2 IASD

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ForgeGAN Group

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Introduction

Generative Models

Generative Adversarial Networks (GANs) [1] aim to learn a distribution p_g that approximates the real data distribution p_{data} in order to generate realistic samples.

Challenges

- GAN training can be unstable.
- Sampling methods may create a trade-off between **sample quality** and **diversity**.
- The choice of divergence used to train the discriminator can strongly influence training behavior.

Key Question

Can different f-divergences and sampling strategies improve the quality and diversity of generated samples?

Papers & Implementation

Paper 1: f-GAN

f-GAN [2] proposes a general framework to train Generative Adversarial Networks using different f-divergences between the real data distribution p and the generator distribution q.

Key Idea

- Standard GANs implicitly optimize the Jensen–Shannon divergence.
- f-GAN generalizes this by allowing many divergences.
- Examples: Jensen–Shannon, Kullback–Leibler, Pearson χ^2.

Paper 2: DG$f$low

DG$f$low [3] proposes a post-hoc refinement method that improves generated samples by updating latent vectors using discriminator gradients, without retraining the generator.

f-GAN: Generalizing GAN Divergences

f-Divergence

An f-divergence measures the discrepancy between two probability distributions p and q:

$$D_f(p\|q) = \mathbb{E}_{x\sim q}[f(u)]$$

where f is a convex function and $u = \frac{p(x)}{q(x)}$ is the density ratio.

Examples of f-divergences

Divergence	**Generator** $f(u)$
KL	$u \log u$
Jensen–Shannon	$u \log u - (u+1) \log \frac{u+1}{2}$
Pearson χ^2	$(u-1)^2$

Key Idea

Different choices of f lead to different gradient behaviors during training.

f-GAN Training Principle

Variational Formulation

The f-GAN paper shows that f-divergences can be optimized through a variational objective:

$$D_f(p\|q) = \sup_T \left(\mathbb{E}_{x\sim p}[T(x)] - \mathbb{E}_{x\sim q}[f^*(T(x))]\right)$$

where $T(x)$ is implemented by the discriminator.

GAN Interpretation

- The discriminator estimates the variational function $T(x)$
- The generator updates its samples to minimize the divergence
- Training therefore approximates $p_g \approx p_{data}$

Why it Matters

Changing the divergence changes the training dynamics and the gradients guiding the generator.

Motivation

Even after training, the generator may produce samples that do not perfectly lie on the data manifold. Standard sampling strategies modify the latent distribution globally, but they cannot correct individual samples.

Standard Sampling Strategies

- **Normal sampling:** draw $z \sim \mathcal{N}(0, I)$ without any correction.
- **Soft truncation [4]:** smoothly bias samples toward the center of the latent distribution.
- **Hard truncation [5]:** clip latent values beyond a threshold, improving quality but often reducing diversity.

DG$f$low Idea

Instead of modifying the latent distribution globally, DG$f$low refines each latent vector individually using discriminator gradients, allowing sample-specific corrections.

Latent Update Rule

DG$f$low refines latent vectors by following gradients derived from the f-divergence:

$$z_{t+1} = z_t - \eta \nabla_z f'(r(x)), \quad x = G(z_t)$$

where

$$r(x) = \frac{p_{data}(x)}{p_g(x)}$$

is the density ratio between real and generated data.

Why DG$f$low Can Improve Samples

- The discriminator provides information about how realistic a sample is.
- Latent vectors are moved toward regions producing higher-quality samples.
- Unlike truncation, the refinement is **adaptive and sample-specific**.

Practical Implementation

In practice, the density ratio is not directly accessible and is approximated using the discriminator output.

Our Adaptation of the Method

Goal

We study how different f-divergences interact with DG$f$low refinement and how sampling strategies influence the quality and diversity of generated samples.

Tsested Models

- VanillaGAN (baseline)
- f-GAN with Jensen–Shannon divergence
- f-GAN with Kullback–Leibler divergence
- f-GAN with Pearson χ^2 divergence

Why These Divergences?

The original f-GAN paper shows that JS, KL and Pearson are among the most representative and commonly used f-divergences, offering different gradient behaviors during training.

Two Discriminator Architectures

- **VanillaGAN**: Fully connected 1024 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 1, Leaky ReLU, **dropout added (p=0.3) for stabilization**.
- **f-GAN**: Same layer sizes, Leaky ReLU, **spectral normalization [6] applied for stability**, output as logits for variational function.

Divergence-Specific Implementation

- Each divergence (JS, KL, Pearson) requires specific f and f^* functions.
- JS uses special activation to keep outputs numerically stable.

Training Hyperparameters

- Batch size: 64 (training), 2048 (generation), latent dim: 100
- Number of epochs: 600
- Adam optimizer with divergence-specific learning rates: D: 1e-4 (JS), 2e-4 (KL/Pearson), G: 2e-4 for all
- Gradient clipping for stability
- Leaky ReLU in hidden layers, optional sigmoid on output

Sampling Strategies Compared

- Normal Gaussian sampling
- Soft truncation
- Hard truncation
- DG$f$low refinement (sample-specific corrections using discriminator gradients)

DG$f$low Refinement Details

- Latent vectors refined iteratively, generator frozen
- Gaussian noise added ($\gamma = 0.01$) to preserve diversity
- Step size η adapted for stability: 0.1 for JS/KL, 0.001 for Pearson

Why DG$f$low Outperforms Other Sampling Methods

- Normal, soft truncation, and hard truncation treat all samples equally
- DG$f$low allows sample-specific improvements
- Pearson requires smaller step size to avoid unstable latent updates

Results & Interpretation

Model	Sampling	FID	Precision	Recall
VanillaGAN	Normal	0.0208	0.7338	0.8256
	Soft Truncation	0.0664	0.8264	0.6528
	Hard Truncation	0.4905	0.8938	0.2015
	DG$f$low	0.0207	0.7460	0.8270
f-GAN JS	Normal	0.0059	0.8025	0.7935
	Soft Truncation	0.0384	0.8730	0.6227
	Hard Truncation	0.2592	0.8780	0.2496
	DG$f$low	0.0056	0.8417	0.7504
f-GAN KL	Normal	0.0080	0.7524	0.8211
	Soft Truncation	0.0283	0.8316	0.6635
	Hard Truncation	0.2399	0.8306	0.2903
	DG$f$low	0.0106	0.7913	0.7328
f-GAN Pearson	Normal	0.0059	0.7909	0.7868
	Soft Truncation	0.0264	0.8537	0.6497
	Hard Truncation	0.2184	0.8663	0.2385
	DG$f$low	0.0060	0.8127	0.7879

Table 1: Metrics for local experiments with different models and sampling strategies.

Model	Sampling	FID	Precision	Recall
f-GAN JS	Normal	8.51	0.86	0.64
	DG$f$low	7.38	0.87	0.62
f-GAN KL	Normal	11.62	0.84	0.66
	DG$f$low	10.56	0.84	0.62
f-GAN Pearson	Normal	9.78	0.84	0.66
	DG$f$low	9.30	0.85	0.65

Table 2: Leaderboard metrics obtained from the platform for submitted models.

Main Observations

- DG$f$low consistently improves or matches FID and recall compared to Normal sampling.
- Soft and Hard truncation rarely outperform Normal or DG$f$low.
- JS and KL divergences yield the best FID $\rightarrow$ higher sample quality.
- Pearson is sensitive to step size η for DG$f$low.
- Platform results confirm trends: DG$f$low improves FID for JS, KL, and Pearson, while precision and recall remain similar.

Key Limitations

- DG$f$low requires careful η tuning, especially for Pearson.
- Training KL and Pearson f-GANs is sensitive to gradients, even with clipping.
- Large batch sizes needed for reliable FID/Precision/Recall $\rightarrow$ higher computation.
- Results may vary between local tests and platform evaluations.
- Generalization to larger datasets or architectures is not guaranteed.

Conclusion

Summary

- Different f-divergences affect discriminator behavior and gradient stability; JS and KL provide smooth guidance, Pearson requires careful step size tuning.
- Sampling strategies impact the precision–recall trade-off: Hard/Soft truncation reduce diversity, while DG$f$low refines individual samples without globally restricting the latent space.
- DG$f$low improves FID across the tested divergences, including JS, KL, and Pearson, although Pearson requires more careful tuning.
- Overall, divergence choice and sampling method jointly determine generative performance; refinement offers an additional way to enhance sample quality.

REFERENCES

[1] Ian J. Goodfellow et al. "Generative Adversarial Nets". In: *Advances in Neural Information Processing Systems*. Vol. 27. Curran Associates, Inc., 2014.

[2] Sebastian Nowozin, Botond Cseke, and Ryota Tomioka. "f-GAN: Training Generative Neural Samplers using Variational Divergence Minimization". In: *Advances in Neural Information Processing Systems*. Vol. 29. Curran Associates, Inc., 2016.

[3] Abdul Fatir Ansari, Ming Liang Ang, and Harold Soh. "Refining Deep Generative Models via Discriminator Gradient Flow". In: *International Conference on Learning Representations*. 2021.

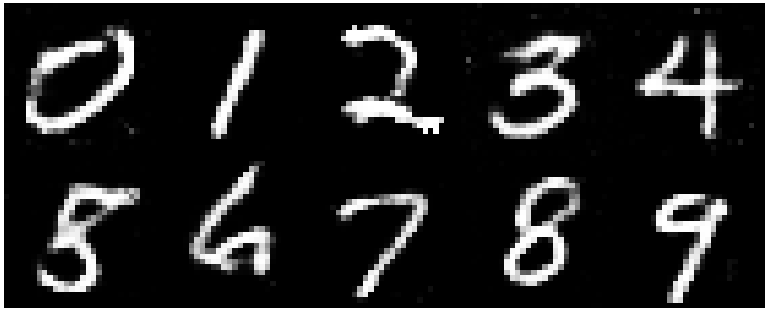
[4] Tuomas Kynkäänniemi et al. "Improved Precision and Recall Metric for Assessing Generative Models". In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc., 2019.

[5] Andrew Brock, Jeff Donahue, and Karen Simonyan. "Large Scale GAN Training for High Fidelity Natural Image Synthesis". In: *International Conference on Learning Representations*. 2019.

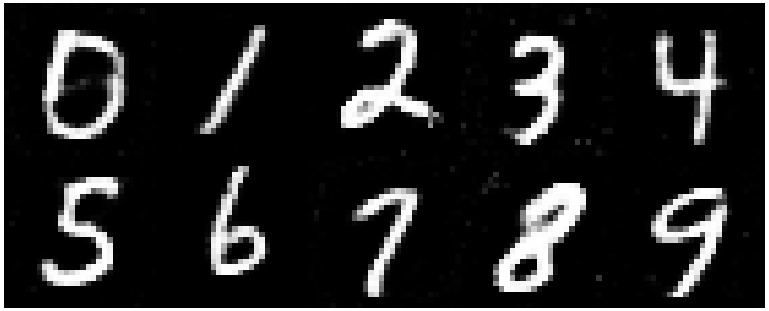
[6] Takeru Miyato et al. "Spectral Normalization for Generative Adversarial Networks". In: *International Conference on Learning Representations*. 2018.

Appendix

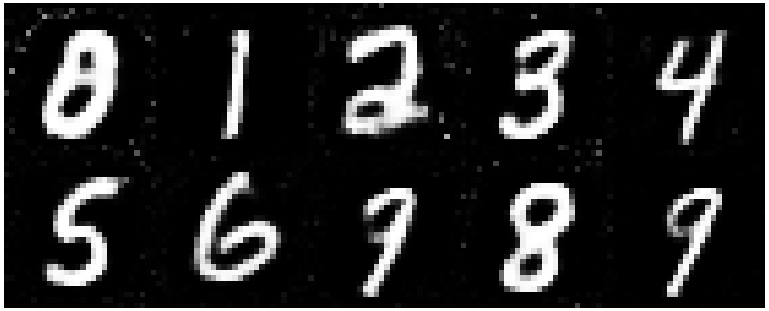
VanillaGAN Generated Images



(a) Normal



(b) Soft Truncation



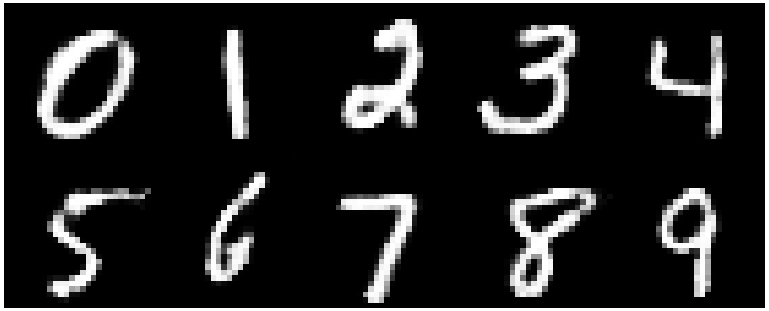
(c) Hard Truncation

(d) DG$f$low

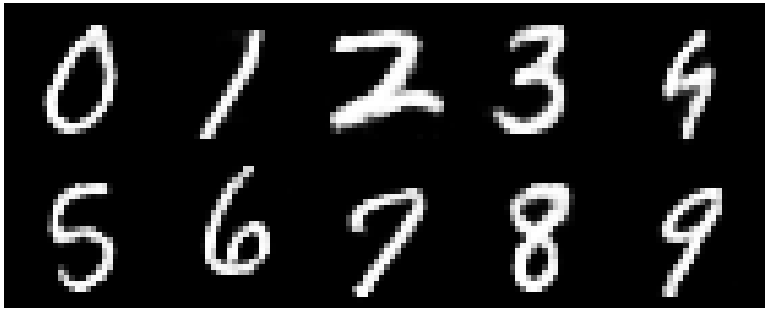
Figure 1: VanillaGAN generated images across different sampling strategies.

f-GAN JS Generated Images

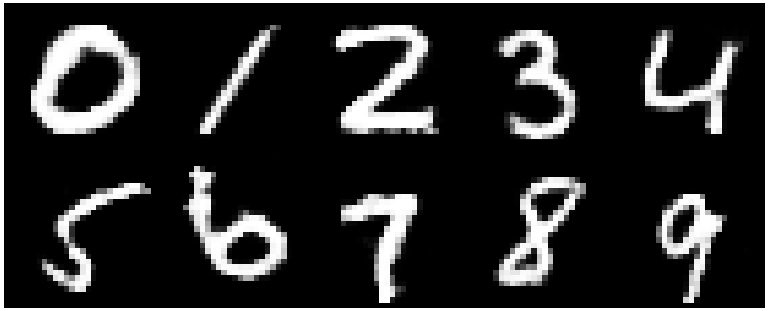
(a) Normal



(b) Soft Truncation



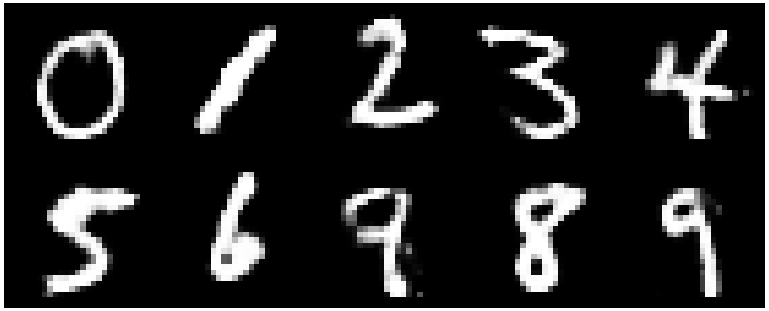
(c) Hard Truncation



(d) DG$f$low

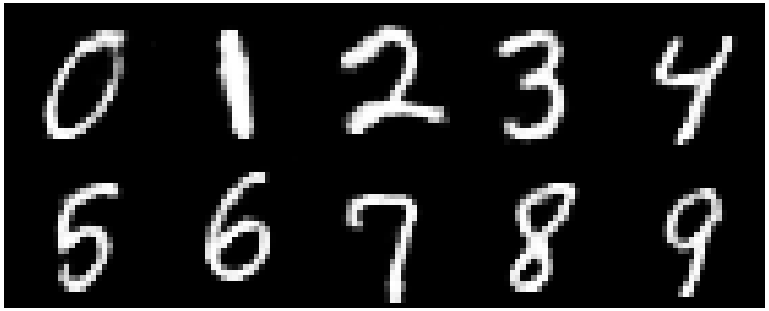
Figure 2: f-GAN JS generated images across different sampling strategies.

f-GAN KL Generated Images

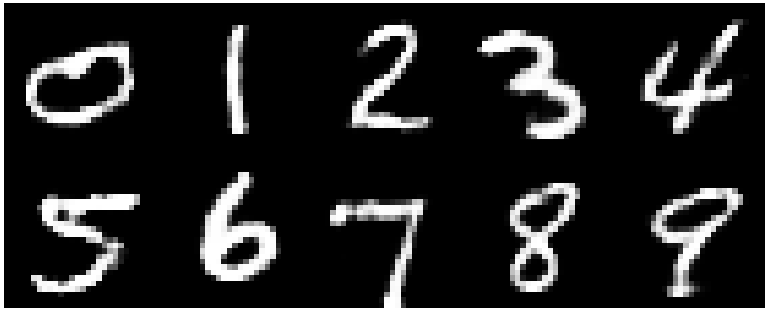


(a) Normal

(b) Soft Truncation



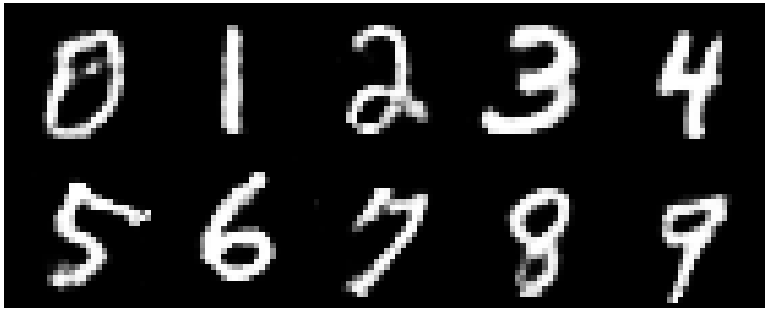
(c) Hard Truncation



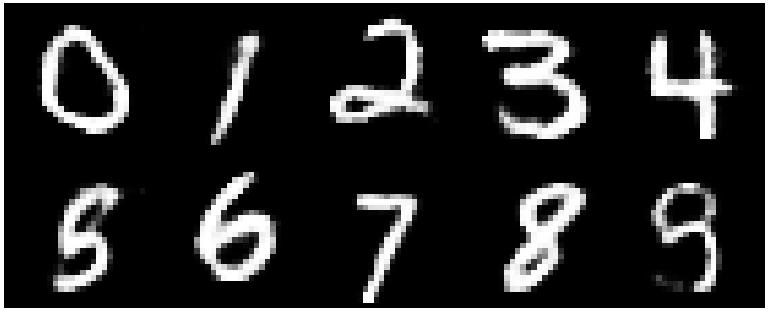
(d) DG$f$low

Figure 3: f-GAN KL generated images across different sampling strategies.

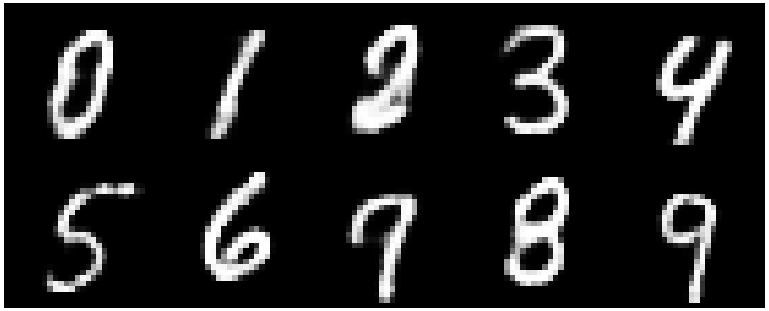
f-GAN Pearson Generated Images



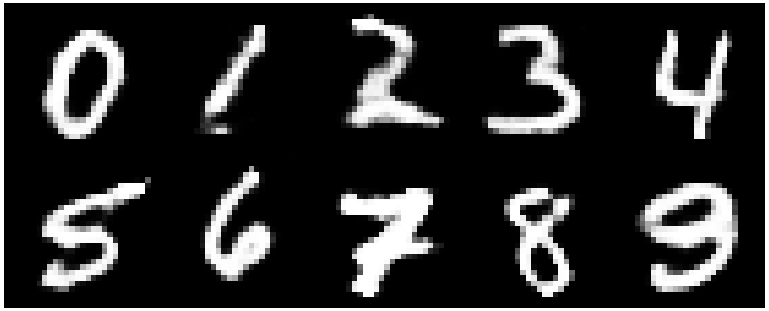
(a) Normal



(b) Soft Truncation



(c) Hard Truncation



(d) DG$f$low

Figure 4: f-GAN Pearson generated images across different sampling strategies.